Worksheet 3 — Your STEM Pathway in Nuclear

High School · Grades 9–12 · Career, College & Self-Assessment · ~40 min

Nuclear energy is one of the most interdisciplinary fields in STEM — it needs physicists, materials scientists, software engineers, electricians, machinists, policy analysts, communicators, and more. This worksheet helps you see where your own strengths and curiosities could lead.

Name: ______________________________ Date: ______________ Class: ______________

Part A · Map the Discipline to the Job

STEM area you enjoy	Nuclear-industry role	Typical degree path	Median US salary (20
Calculus + physics	Reactor / nuclear engineer	BS/MS Nuclear or Mech. Eng.	$125k
Chemistry + materials	Fuel cycle / materials scientist	BS/PhD Mat. Sci. or Chem.	$110k
Computer science	Reactor simulation / digital twin dev	BS CS or Computational Eng.	$130k
Electrical / electronics	I&C engineer, grid integration	BS Electrical Eng.	$115k
Biology + safety	Health physicist, radiation protection	BS/MS Health Physics	$105k
Hands-on / mechanical	Reactor operator, NDT technician	AS + NRC license	$100k
Writing / policy	Regulatory analyst, nonproliferation	BA + MA Policy / Law	$95k
Business / finance	Project finance, utility planning	BS/MBA Finance or Econ.	$120k

Salary figures are rough US estimates compiled from BLS, ANS career data, and industry surveys; actual figures vary by region, employer, and experience.

Part B · Self-Inventory

1. Which two STEM areas above appeal to you most right now? Why?

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2. Which one is the biggest *stretch* for you — and would you be willing to grow into it?

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3. Of the eight roles above, which one would you most want to job-shadow for a day? What three questions would you ask that professional?

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4. Pick a current nuclear company (e.g., Westinghouse, NuScale, X-energy, Helion, Radiant, Terrestrial Energy, Constellation Energy). Visit their careers page and list three open positions plus the

qualifications each requires.

5. Identify one high school class you can take next semester that moves you toward your top choice (AP Physics, AP Chemistry, AP Calculus, AP Computer Science, Engineering Design, etc.).

Part C · Build a Four-Year Plan

Sketch a high school course sequence that prepares you for a nuclear-adjacent STEM major. Add one extracurricular or summer experience per year (robotics, science fair, internship, DOE-NE student programs, ANS Student Conference, etc.).

Year	Math	Science	Tech / Eng	Extracurricular / Summe
9th				
10th				
11th				
12th				

Part D · Resources to Explore

- American Nuclear Society (ANS) student programs — ans.org/students
- U.S. DOE Office of Nuclear Energy — energy.gov/ne
- Nuclear Regulatory Commission Student Programs — nrc.gov
- Nuclear Energy Institute (NEI) Careers — nei.org/careers
- Idaho National Laboratory (INL) internships — inl.gov/careers
- MIT OpenCourseWare 22.01 — Intro to Nuclear Engineering (free)
- edX / Coursera — search 'nuclear energy' for free university courses